

Deep Learning Based Stress Prediction From Offline Signatures

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Abstract—Soft-Biometric measurements are now increasingly adopted as a robust means of determining individual's non-unique characteristics with the emerging models that are widely used in the deep learning domain. This approach is clearly valuable in a variety of scenarios, specially those relating to forensics. In this study, we specifically focus on stress emotion, and propose automatic stress prediction technique from offline signature biometrics using well-known deep learning architectures such as AlexNet, ResNet and DenseNet. Due to the limited number of research that study emotion prediction from offline handwritten signatures with deep learning methods, best to our knowledge this is the first experimental study that presents empirical achievable prediction accuracy around 77%.

Index Terms—offline signature, soft-biometrics, emotion prediction, stress prediction, deep learning

I. INTRODUCTION

Nowadays, the availability and necessity of digitised documents grow rapidly [1], and mainly handwritten signature used by people to sign these documents such as checks, contracts, credit card applications, visa applications and so on. Since the amount and availability of signed documents has increased tremendously, the growth of signature fraud is increased as well [2]. Noting that handwritten signatures provide a rich source of individual's characteristics; it becomes increasingly more important to enhance our ability to analyse, investigate and extract knowledge from such rich sources.

Signing is a writing activity which requires the coordination of muscles and the movements involving the synchronisation of arms, hands and fingers with the eyes [2] [3]. In addition, emotional feelings, health conditions, age, gender, the environment that individual signs, provided signing box for the signature and many other factors play important role during the signing process. All of these factors influence the way that individuals sign. Hence, there is a long history in the literature presenting that forensics benefit from the fields of biometrics by identifying the identity of individuals from the online and/or offline handwritten signatures or detecting the unskilled or skilled forgeries [2] [4] [5] [6] [7].

Problematic issues in forensic applications are not limited to the ones explained above and in some circumstances, they may not be sufficient or even useful. For instance, in some cases the more specific traits are helpful, such as where the determination of “handedness”, “age”, “gender” or “higher

mental status” (i.e. emotion) might be important. Therefore, soft-biometrics also started to play an important role in forensic application areas where reliable and robust identification and/or verification of individual's non-unique characteristics is required.

Soft-biometric research community made very moderate effort, and study age, gender and handedness [4] [8] [9] [10] prediction from handwritten signature. However, it is very difficult to find studies investigating higher mental status (i.e. emotion) prediction from online signature biometrics with the exception of the recent study presented in [11]. On the other hand, there is not any available study in the literature related to the emotion prediction from offline signature. Automatic estimation of emotion from signature biometrics can be a powerful and important tool to support a range of practical scenarios and applications for forensics. For instance, in interpreting particular case or evaluating human activities for forensics, knowledge of whether an individual is relaxed or under stress at a particular time or under any circumstances can provide extremely significant valuable information.

In this paper, we report several experiments to analyse and explore the predictive capability of stress based emotion from offline handwritten signature by adopting a deep learning structure in order to provide a potential solution to a range of important practical problems of feature extraction stage. We believe that presented experimental results and discussions in this study is very significant in the sense that it presents first preliminary evidence that it is possible to predict stress emotion from offline handwritten signatures and shed some more light for further studies to develop more powerful and robust emotion prediction techniques.

II. BACKGROUND

Handwritten signature data can be represented as a spatial (offline) or a temporal (online) information. If signature biometric is acquired through scanner, then it can only be represented as offline information. On the other hand, if signature is acquired through a digital tablet, then it can be represented as both offline and online information. Offline representation is an image containing static information of signature, where the online representation is the collection of data (such as pen pressure, timestamp, x and y coordinates,

pen orientation, etc.) containing the dynamic information of signature.

In the case of forensic applications, handwritten signature biometric data can be obtained from crime scenes as a paper or from digitised tablet as a document. Since, for both of these cases, the handwritten signature biometric data would be represented as an image (offline), in this study we investigated the stress prediction based on offline signature biometric data. As stated in [12], this considered to be more difficult task than the online. In this section, other reported works related to our study and also several other relevant studies related to the online signature and soft-biometric prediction in general will be described in detail. This will allow us to understand the predictive capabilities from handwritten signature biometrics in a wider context.

Techniques that uses soft-biometric information to improve identity prediction accuracy is proposed in study [8]. More specifically, age, gender and handedness soft-biometric information is used as a feature selector and as an extra input feature in identity prediction task. Their encouraging results also presents that the soft-biometrics predictive capabilities from online handwritten signature is very powerful.

More directly, age soft-biometric prediction from online handwritten signature is explored in study [10]. Presented preliminary results were indicating 73.64% prediction accuracy. Although, this is very moderate accuracy, it was accepted as extremely positive, since this area has not yet been extensively investigated to date.

Gender prediction from Hindi handwritten signature using different feature sets (including and not including Euler number) and classification techniques such as Support vector Machine(SVM) and Back Propagation Neural Network (BPNN) have been investigated in [13]. Proposed method were able to achieve gender prediction accuracy around 80%.

More recently, emotion prediction (specifically happy, sad and stressed) from online handwritten signature is analysed using three classification techniques namely K-Nearest Neighbor (k-NN), JRIP and Random Forest in [11]. Presented results shows that according to the experimental methodology (i.e. dividing the classes into sad/not sad, happy/not happy and stressed/not stress based on emotion scores and/or different tasks) performance of prediction accuracy can change from 60% to 99%.

As a summary, there is not a considerable amount of efforts in the field of soft-biometric in general related to the handwritten signature biometrics. However, as can be noticed from the above discussion there is not any available study related to the emotion prediction from offline handwritten signature biometrics.

The main reason behind this is that all publicly or commercially available databases do not include individuals demographic labels or emotional status in addition to their identification labels. Recently, a new database [11] is published which presents online and offline handwritten signature biometric samples with a soft-biometric labels. Hence, it is timely and important to explore predictive capabilities of emotion from

the handwritten biometric data for forensic applications as discussed in the previous section.

In this paper, a particular goal is conducting a detailed preliminary study to predict stressed emotion from handwritten signature using deep learning methods, since deep learning approaches are recently being more popular and increasingly adopted for both identity [14] and soft-biometric [13] prediction by the biometrics research community.

III. METHODOLOGY

Our experimental methodology is defined based on the significant issues reported in the previous section. In general, the stress prediction methodology in this study encompasses a stage of definition of offline signature handwritten biometric data, pre-processing of signature data followed by prediction of stress emotion using deep learning approaches. Each of these stages will be explained in detail in the following subsections.

A. Database

A recently published, proposed and analysed handwriting and signature database [11] is utilised for our experimental study.

The acquisition of handwritten signature samples in this database was carried out in a homogeneous environment using a graphics tablet (Wacom Intuos Pro-large large and a special writing device called Intuos Inkpen (Wacom Art Pen - KP701E2)) with a sample rate of 100 Hz, under the guidance of a supervisor.

The database consists of 134 individuals and each individual generated 15 signatures in each of four different sessions (total of 60 signatures per individual). As a total database contains 8040 handwritten signature samples.

In session 1, individuals directly allowed to sign. In session 2 and 3, initially individuals watch a video which conveys a “positive” and “sad” message respectively and then allowed to sign. In session 4, initially individuals allowed to write a text within a limited time (using a countdown timer) which conveys “stress”, and then allowed to sign. After each of these session, individuals are allowed to express their feelings by filling in the score sheet, scoring their emotional states namely happy, sad and stressed by assigning a score from 1 (least) to 10 (most).

B. Pre-processing

Initially, the signature database described in the previous subsection, where the offline handwritten signature biometric samples have the stress emotion label are utilised to form an experimental dataset.

In this database, for each subject at each session, signatures collected in an equally divided 15 boxes on A4 paper and each signature sample saved as a separate image. Thus, each signature appears at different locations in the image as shown in the Figure 1. In addition, each individual signed their signature at different locations in these boxes. Hence, in order to crop and obtain only the signature region in each image,

segmentation is applied. It was very hard to segment signatures from the images by defining equally distributed 15 boxes. Since, all the signature biometric data are collected using a blue colored pen, color based segmentation is applied to extract signatures more easily and efficiently as shown in the Figure 1.

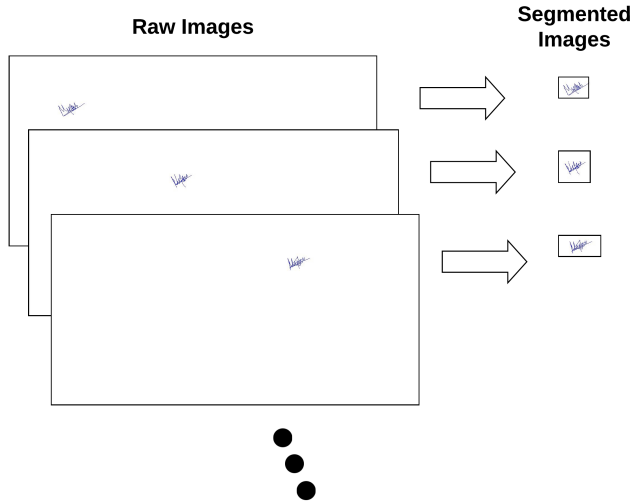


Fig. 1. Segmentation of the Signatures from Raw Image

After forming a dataset of 8040 handwritten signature images, we define the class labels by using the ground truth (scale from 1 to 10) - described in Section III A - provided for the stress level by means of simple thresholding. All the signatures which are associated with level four or less stress are labeled as "Neutral" and the rest is labeled as "Stressed". We acknowledge, this is to an extent subjective. However, level four stress is selected as a cutoff point, mainly to maintain the equivalent distribution of the data to the given classes.

C. Prediction

Following the dataset formation and pre-processing of the handwritten signature biometric data, stress prediction is performed using three well-known and widely used deep learning models namely, AlexNet [15], ResNet [16] and DenseNet [17].

Models selected for this study are taken from PyTorch framework's models module. For ResNet, Resnet50 and for DenseNet, DenseNet121 architectures are used.

To evaluate the models, holdout method with 80:20 train:test ratio is applied to five times to different train and test sets generated by shuffling the dataset five times and applying split operation with the specified ratio.

Overall flow of the system including database, pre-processing and prediction is presented in Figure 2.

IV. RESULTS AND DISCUSSIONS

Stress emotion prediction performance is evaluated and analysed by using the step by step methodology defined in the previous section; using all signature biometric data from

all sessions, using two class labels (neutral and stressed) and different deep learning models. The results of each experiment is presented in Table I.

TABLE I
RESULTS

	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	Avg
Resnet50	51.12	54.88	58.31	57.13	55.75	55.44
DenseNet	67.13	73.38	61.07	65.94	70.25	67.55
AlexNet	77.44	77.07	75.63	77.06	75.88	76.61

The attainable accuracy of stress prediction from offline handwritten signature data varies as 55%, 68%, 77% with respect to deep learning models of ResNet, DenseNet and AlexNet.

Reasons behind these different achieved performances generated by deep learning models can be explained as follows; AlexNet is one of the earliest deep learning architectures (2012) proposed for object recognition tasks, which has been designed with limited amount of data and limited computational complexity compared to current resources. Hence, AlexNet architecture has a simple layers with large number of feature maps which can successfully extract unique features from handwritten signature biometric samples. On the other hand, DenseNet architecture has a smaller number of filters in its convolutional layers, which for the particular case here, may result in a lack of unique features to predict stress from signature biometric samples. In contrast to DenseNet and AlexNet, ResNet has multiple numbers of convolutional layers and feature maps. However, similarly to the DenseNet, for the prediction task presented in this study, due to its deepness, it might focus on more deeper features that might not be unique for the prediction of stress.

Since, there is not any formally documented work which directly achieves successful prediction of emotion/mental state from offline handwritten signature data, it is not possible to compare our results.

However, we can try to compare our stress prediction results with the recently presented work in [11]. It is quite hard to make a bias comparison since there are many different aspects in terms of adopted methodology. Lets perform the comparison by taking into account each of these differences;

- Representation of signatures: offline and online signature data is adopted in this paper and in [11] respectively. As mentioned previously, it is more harder to obtain unique features from offline data rather than online [12].
- Dataset: in this paper, all signature data (collected under neutral, happy, sad and stress conditions) is included, while in [12], only signature data from two sessions (collected under neutral and stress conditions) are included. Since, the formed dataset in this paper contains handwritten signature samples signed under more varying emotional conditions, prediction task became more challenging.
- Class Labels: score value of 4 and 8 is used as a threshold to define class labels in this study and in [11] respectively.

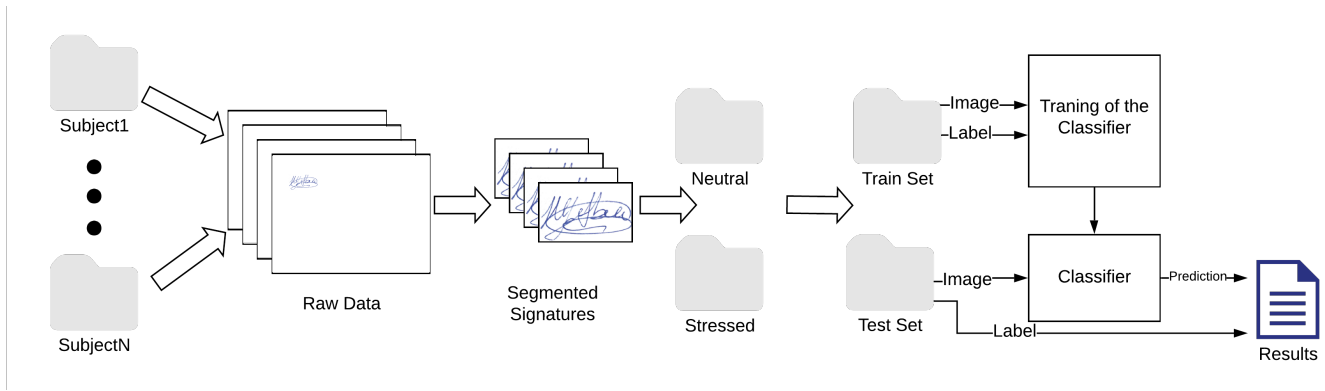


Fig. 2. Flow of the Proposed System

Decreasing the threshold again makes the prediction task more challenging for our study.

- Feature Extraction: in this work, feature extraction is done by the models' convolutional layers, hence there are no handcrafted features, while in [11] several dynamic and static features are extracted from online signature samples. Here, our methodology is more advantageous, since extracting features from offline signature data (i.e. from image) is very complicated task [12].
- Prediction: in this paper, deep learning methods adopted while in [11] K-Nearest Neighbor, JRIP and Random Forest classifiers are adopted for the prediction task.
- Results: stress prediction performance varies from 55%-77% and from 65%-72% in this paper and in [11] respectively. These results are very comparable.

Although, achieved stress prediction performance are similar, it is very difficult to draw completely firm conclusions from what we are comparing under these discussed different circumstances. However, the results presented nevertheless raise some very interesting issues and possibilities. For instance, there is evidence that it is possible to identify the "higher level" state, specifically reflecting the stress emotion of an individual, based on individual's offline signature. In addition, it can be noted that the predictive capability will depend to an extent on the choice of classification infrastructure (i.e. deep learning models).

V. CONCLUSION

In this paper, we have presented an experimental analysis for predicting higher level states, specifically stress emotion, from biometric measurements based on offline handwritten signature, suggesting why this might provide a convenient and appropriate framework to support forensic applications.

Two principal novel points emerge here. First, these are the first reported results to show the reliable prediction of stress emotion from the offline (i.e. image) signature biometric sample of an individual. Second, we have shown how adopting deep learning structure offers substantial benefits in this type of scenario, providing potential solutions to a range of important practical problems of feature extraction stage.

This study, in general, demonstrates the potential of the proposed methodology by using a large dataset of signature samples collected from a diverse range of subjects and under different emotional conditions, and represents perhaps a first step in stimulating a more focused approach to developing this area.

REFERENCES

- [1] G. Saha, "Design digital signature system for paperless operation," in *2017 International Conference on Communication and Signal Processing (ICCSP)*, April 2017, pp. 0324–0328.
- [2] S. J. Gideon, A. Kandulna, A. A. Kujur, A. Diana, and K. Raimond, "Handwritten signature forgery detection using convolutional neural networks," *Procedia Computer Science*, vol. 143, pp. 978–987, 2018.
- [3] A. K. Jain, A. A. Ross, and K. Nandakumar, *Introduction to Biometrics*. Springer Publishing Company, Incorporated, 2011.
- [4] T. Chatzisterkotis, "An examination of quantitative methods for forensic signature analysis and the admissibility of signature verification system as legal evidence." Ph.D. dissertation, University of Kent, May 2015. [Online]. Available: <https://kar.kent.ac.uk/54048/>
- [5] R. Tolosana, R. Vera-Rodriguez, J. Fierrez, and J. Ortega-Garcia, "Exploring recurrent neural networks for on-line handwritten signature biometrics," *IEEE Access*, vol. 6, pp. 5128–5138, 2018.
- [6] R. Sanchez-Reillo, H. C. Quiros-Sandoval, I. Goicoechea-Telleria, and W. Ponce-Hernandez, "Improving presentation attack detection in dynamic handwritten signature biometrics," *IEEE Access*, vol. 5, pp. 20 463–20 469, 2017.
- [7] M. Liwicki, C. E. v. d. Heuvel, B. Found, and M. I. Malik, "Forensic signature verification competition 4nsigcomp2010 - detection of simulated and disguised signatures," in *2010 12th International Conference on Frontiers in Handwriting Recognition*, Nov 2010, pp. 715–720.
- [8] M. Abreu and M. Fairhurst, "Improving identity prediction in signature-based unimodal systems using soft biometrics," in *Biometric ID Management and Multimodal Communication*, J. Fierrez, J. Ortega-Garcia, A. Esposito, A. Drygajlo, and M. Faundez-Zanuy, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 348–356.
- [9] M. d. Costa-Abreu and M. Fairhurst, "Improving handwritten signature-based identity prediction through the integration of fuzzy soft-biometric data," in *2012 International Conference on Frontiers in Handwriting Recognition*, Sep. 2012, pp. 797–802.
- [10] M. Erbilek, M. Fairhurst, and M. Da Costa-Abreu, "Improved age prediction from biometric data using multimodal configurations," in *2014 International Conference of the Biometrics Special Interest Group (BIOSIG)*, Sep. 2014, pp. 1–7.
- [11] Y. B. Ayzeren, M. Erbilek, and E. Celebi, "Emotional state prediction from online handwriting and signature biometrics," *IEEE Access*, vol. 7, pp. 164 759–164 774, 2019.
- [12] M. Liwicki, M. I. Malik, C. E. v. d. Heuvel, X. Chen, C. Berger, R. Stoel, M. Blumenstein, and B. Found, "Signature verification competition for online and offline skilled forgeries (sigcomp2011)," in *2011 Interna-*

tional Conference on Document Analysis and Recognition, Sep. 2011, pp. 1480–1484.

- [13] Y. Sun, F. P. . Lo, and B. Lo, “A deep learning approach on gender and age recognition using a single inertial sensor,” in *2019 IEEE 16th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, May 2019, pp. 1–4.
- [14] K. Sundararajan and D. L. Woodard, “Deep learning for biometrics: A survey,” *ACM Comput. Surv.*, vol. 51, no. 3, May 2018. [Online]. Available: <https://doi.org/10.1145/3190618>
- [15] A. Krizhevsky, “One weird trick for parallelizing convolutional neural networks,” *arXiv preprint arXiv:1404.5997*, 2014.
- [16] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [17] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4700–4708.
- [18] A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, “Automatic differentiation in pytorch,” in *NIPS-W*, 2017.